Noah A

Principal Data Engineer | Senior Data Engineer | Data Engineer

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PROFILE

Senior / Lead Data Engineer with 11+ years of experience designing, building, scaling, and leading modern data platforms in startup and high-growth environments. Deep expertise in real-time streaming, cloud-native data architectures, and large-scale ETL/ELT pipelines using SQL, Python, Spark, Kafka, and Lakehouse frameworks across AWS, Azure, and GCP. Proven leader with hands-on experience mentoring engineers, owning end-to-end data architecture, and driving engineering best practices. Adept at partnering with product, analytics, and business stakeholders to translate complex requirements into resilient, high-performance data solutions that enable analytics, AI/ML, and data-driven decision-making.

SKILLS

Data Engineering & Architecture
- Batch & real-time data pipelines, Lambda architecture, Kappa architecture, Lakehouse architecture, data lakes, data warehouses, data marts, cloud data integration

Programming & Querying
- SQL, advanced SQL, CTEs, window functions, query optimization, Python, ETL, ELT, data transformation, automation, Java, Scala, Bash, R

Databases & Storage
- SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, Cassandra, Redis, DynamoDB, SQLite, InfluxDB, TimescaleDB, Parquet, ORC, Delta Lake

Cloud Platforms
- AWS (S3, EC2, EMR, Glue, Lambda, Kinesis), Azure (Data Factory, Blob Storage, HDInsight), GCP (Dataflow, Pub/Sub), Snowflake, Databricks, Azure Synapse, BigQuery

Data Quality, Governance & Security
- Data quality management, metadata management, data lineage, master data management, data governance, data privacy, GDPR, HIPAA

Domain & Leadership
- Healthcare data (EHR, claims, HL7), IoT analytics, predictive maintenance, supply chain optimization, technical leadership, mentoring, architecture ownership, Agile/Scrum, Jira, Confluence

Streaming & Big Data
- Apache Kafka, Kafka Streams, Kafka Connect, Apache Flink, Apache Spark, Spark Streaming, Hadoop, Hive, HBase, Presto, Apache Storm, AWS Kinesis

Data Modeling & Warehousing
- Star schema, Snowflake schema, dimensional modeling, fact-dimension modeling, Data Vault, Kimball methodology, ER diagrams, schema design, UML

ETL & Orchestration Tools
- Apache Airflow, Apache NiFi, dbt, Talend, Informatica, Pentaho, SSIS, Alteryx, ETL optimization, data migration, real-time data processing

DevOps & DataOps
- Git, CI/CD, Terraform, CloudFormation, Pulumi, Docker, Kubernetes, Jenkins, GitLab CI, CircleCI, Ansible, automated pipeline testing, data monitoring

Analytics, ML & Visualization
- Spark MLlib, ML data pipelines, feature engineering, model deployment, MLflow, TensorFlow, Scikit-learn, PyTorch, Tableau, Power BI, QuickSight, Jupyter Notebooks

PROFESSIONAL EXPERIENCE

Falkonry 08/2021 – Present
Principal Data Integration Engineer

- Designed and implemented real-time streaming pipelines using Apache Kafka, Kafka Connect, and Apache Flink to support live inventory, supply-chain, and operational analytics, enabling high-frequency monitoring of critical KPIs.
- Built scalable Spark and Databricks workloads for processing high-volume IoT and supply-chain datasets, including batch and micro-batch ETL pipelines for reliable data transformation and aggregation.
- Developed Python-based predictive models and ML pipelines to forecast demand, optimize replenishment strategies, and reduce stock-outs by 20%, integrating time-series analytics and feature engineering.

- Architected and maintained a Snowflake-based enterprise data warehouse with star and snowflake schemas, integrating multiple operational and external data sources for unified analytics and dashboard reporting.
- Implemented a Lakehouse architecture supporting both analytical and operational workloads, combining historical data with streaming events for near real-time decision-making and ML readiness.
- Designed time-series data solutions using InfluxDB to monitor waste generation, operational efficiency, and trend analysis, enabling actionable insights and process optimization.
- Automated cloud infrastructure provisioning, scaling, and monitoring using Terraform and AWS services (S3, EMR, Glue, Lambda), ensuring reliable and cost-efficient production environments.
- Provided technical leadership, architecture ownership, and cross-functional collaboration with product and operations teams, defining best practices, mentoring engineers, and delivering data solutions aligned with strategic business KPIs.

Current Health 05/2018 – 07/2021

Data Engineering Team Lead

- Led the design and development of a cloud-native health data platform ingesting real-time streaming data from wearable devices, enabling scalable analytics and personalized user insights.
- Built Kafka-based streaming pipelines and Spark batch processes to ingest, clean, and transform large-scale fitness and health data, ensuring low-latency, reliable data delivery for downstream analytics.
- Implemented data lakes on Google Cloud Platform (GCS, Dataflow, BigQuery) to support both analytical workloads and machine learning pipelines, providing a single source of truth for health metrics.
- Designed and optimized analytical schemas (Star and Snowflake) to support reporting, dashboards, and ML features, improving query performance and enabling near real-time insights into user behavior.
- Developed advanced SQL queries and dashboards to track fitness, activity, and health trends, supporting product and business teams with actionable insights for personalized recommendations.
- Applied Spark MLlib to build and deploy ML models for personalized workout and nutrition recommendations, integrating predictive analytics into the data platform for end-user value.
- Automated data ingestion, synchronization, and orchestration pipelines using Apache NiFi and Python scripts, improving operational efficiency and reducing manual interventions.
- Mentored and led a team of data engineers, establishing coding standards, performance best practices, and collaborative workflows to deliver reliable, production-grade data solutions aligned with business goals.

Seeq Corporation 02/2015 – 04/2018

Data Engineer

- Built a real-time document indexing and retrieval platform using Apache Kafka, Kafka Connect, and Apache Flink, enabling high-throughput processing and near-instant search capabilities.
- Designed scalable storage solutions using Parquet and cloud object storage (AWS S3, Azure Blob) for large document datasets, ensuring cost-efficient and high-performance data access.
- Developed Python-based ETL pipelines to integrate third-party systems, metadata sources, and structured/unstructured data for analytics and reporting.
- Implemented OCR pipelines to convert scanned documents into searchable, structured formats, supporting document analytics and operational workflows.
- Designed and deployed cloud infrastructure on AWS and Azure using Infrastructure-as-Code (Terraform, CloudFormation, Pulumi) for automated, repeatable, and scalable environments.
- Delivered real-time collaboration analytics using Apache Spark to measure usage, track productivity patterns, and provide actionable insights for product and business teams.

KEY PROJECTS

Real-Time Inventory & Supply Chain Analytics Platform
End-to-end platform using Kafka, Flink, and Databricks for real-time inventory and supply chain event tracking. Included scalable Spark pipelines for high-volume event processing, Snowflake star schemas for dashboards, and Python predictive modeling to reduce stock-outs by 20%.

IoT-Driven Waste Tracking & Optimization System
Real-time IoT ingestion and processing system with Kafka and Flink to monitor waste generation and routing efficiency. Leveraged InfluxDB for time-series trend analysis and a Lakehouse architecture to unify streaming analytics with historical reporting.

Cloud-Native Health & Fitness Data Platform
Platform ingesting real-time wearable device data via Kafka streams, with Python and Spark ETL pipelines for data aggregation. Built on Google Cloud Platform with optimized SQL analytics and supported ML-driven personalization using Spark MLlib.

Intelligent Document Management & Search Platform
Real-time document ingestion, indexing, and search system using Kafka and Flink. Designed efficient Parquet-based cloud storage, integrated OCR outputs via Python ETL, and delivered real-time collaboration analytics using Spark.

EDUCATION

Bachelor of Science in Computer Science